

Arrival and required times

Once the graph is levelized, you propagate tags

Goldens to remember

- Forward: $A(\text{in})=0$, $A(u1/Y)=1.2$, $A(\text{out})=3.2$
- Backward setup: $R(\text{out})=10$, $R(\text{in})=6.8$
- Keep these numbers handy, the browser challenges and Track A tests use the same instance

Pseudocode

- Arrival and required need two written passes
- Forward topo takes a max over predecessors for arrival
- Reverse topo takes a min over successors for setup required from the period at sinks
- Open this module's examples file and find the Pseudocode section
- That written sketch is what you implement on the implement track and what the browser

Algorithm sketch

- Goldens on the chain
- Arrival is latest; required is earliest

Algorithm sketch — try these

```
INPUT: DAG G, period, arrival seeds
OUTPUT: A[], R_setup[]
for p in topo(G):
    A[p] ← max over u→p of A[u]+delay    (sources: seed/0)
for sinks: R[p] ← period×cycles
for p in reverse_topo:
    R[p] ← min over p→v of R[v]-delay
GOLDEN: A[out]=3.2; R[out]=10; R[in]=6.8
```

Seed arrival at the source

ARRIVAL AND REQUIRED TIMES

Seed arrival at the source

Step 1 / 5 · seed-arrival · module01-03-arrival-required



Explanation

Set arrival at in to zero for the launch edge. Every other pin waits for its predecessors.

- Arrival at sources starts the wavefront
- Here $A(\text{in}) = 0$
- No delay yet on the first net

$$A(\text{in}) = 0$$

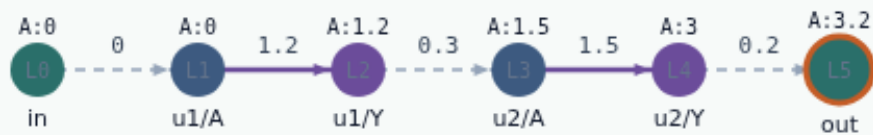
Seed arrival at the source

Propagate arrival forward

ARRIVAL AND REQUIRED TIMES

Propagate arrival forward

Step 2 / 5 · forward-wave · module01-03-arrival-required



Explanation

For each pin in topo order, arrival is the max over predecessors of $A(\text{pred}) + \text{delay}$. At out the wavefront reaches 3.2.

- $A(v) = \max(A(u) + d(u \rightarrow v))$
- Cell delays dominate this toy path
- Golden $A(\text{out}) = 3.2$

$A(u1/Y) = 1.2$
 $A(u2/A) = 1.5$
 $A(u2/Y) = 3.0$
 $A(\text{out}) = 3.2$

Propagate arrival forward

Seed required at the sink

ARRIVAL AND REQUIRED TIMES

Seed required at the sink

Step 3 / 5 · seed-required · module01-03-arrival-required



Explanation

For a single-cycle setup check, required at out equals the clock period—here 10.

- Setup capture edge sets $R(\text{out})$
- $R(\text{out}) = \text{period} \times \text{cycles}$
- Default cycles = 1 $\rightarrow$ 10

period = 10
 $R(\text{out}) = 10$

Seed required at the sink

Propagate required backward

ARRIVAL AND REQUIRED TIMES

Propagate required backward

Step 4 / 5 · backward-wave · module01-03-arrival-required



Explanation

Walk reverse topo order. Required at a pin is the min over successors of $R(\text{succ}) - \text{delay}$. At in, required becomes 6.8.

- $R(u) = \min(R(v) - d(u \rightarrow v))$
- Tightest successor wins
- Golden $R(\text{in}) = 6.8$

$R(u2/Y) = 9.8$
 $R(u1/Y) = 8.0$
 $R(u1/A) = 6.8$
 $R(\text{in}) = 6.8$

Propagate required backward

Keep both tags on every pin

ARRIVAL AND REQUIRED TIMES

Keep both tags on every pin

Step 5 / 5 · both-tags · module01-03-arrival-required



Explanation

Arrival and required live together. Slack at a pin is $R - A$ for setup. Next labs turn those tags into slack and a critical path.

- Forward then backward
- Same graph, two tag maps
- $A(\text{out})=3.2$ and $R(\text{out})=10$

$A(\text{out})=3.2$
 $R(\text{out})=10$
next: slack = 6.8

Keep both tags on every pin

Browser lab track

- In the browser lab, open **arrival-required**
- Load the starter, run the analysis once, and read the metrics panel
- Orient yourself, challenge panel, canvas, Check, then mirror the same goldens in code

Implement track

- In the implement track
- Run ``python3 common/test_propagate.py`` (and the timing-graph test) until the goldens print

Pitfall

- Do not mix setup and hold required maps
- Do not propagate before the graph is levelized
- After an edit or exception, recompute, stale tags lie

Your turn

- Finish the checklist on at least one track, preferably both
- When your numbers match the goldens, take the quiz, then continue